Xi Li

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Profile

Ph.D. researcher in Computer Science at UC Irvine developing multimodal machine learning for computational pathology and single-cell genomics. First author of **STMoE** and **sMMC-22M**, and co-first author of **MUSE**. Research connects histology, spatial gene expression, and biological context through multi-scale modeling, contrastive alignment, and large-scale cell-aligned data. Also fine-tunes and evaluates biomedical foundation models across population/context groups, aiming to make predictions more reliable under distribution shift; contributes to interpretable target discovery through **TargetSage**.

Education

University of California, Irvine — *Ph.D., Computer Science* — Sep 2023 – Present
Tsinghua University — *M.S., Computer Science and Technology* — Sep 2019 – Jun 2022
University of Science and Technology of China — *B.S., Information Security* — Sep 2015 – Jun 2019
Expected Ph.D. graduation: December 2027.

Research Experience

University of California, Irvine — **Irvine, CA**
Ph.D. Student Researcher — *Sep 2023 – Present*

- **STMoE — histology-to-expression learning (first author).** Led development of a multi-scale mixture-of-experts model for **single-cell gene-expression prediction across 25 organs**. Combined single-cell and tissue-context crops, frozen UNI2-h representations, soft contrastive alignment, and organ-routed retrieval with training-only molecular reference banks. Retained **70% of predictive performance** under cross-sample generalization.
- **sMMC-22M — multimodal data and benchmarks (first author).** Built a context-aware resource connecting cell morphology, molecular profiles, and tissue context. The **expanded collection contains 28.3M cell-aligned targets across 25 tissue categories**. Tracked study, assay, and coordinate provenance, distinguishing native Xenium cells from derived Visium HD targets; the paper retains the frozen-release name sMMC-22M.
- **Foundation-model bias and cross-context reliability (first author).** Fine-tuned and evaluated multiple biomedical foundation models across population and biological-context groups. Developed support-calibrated evaluation of aggregate and worst-context performance, **aiming to reduce bias and improve consistency across groups**. Manuscript: *Context Is Not a Side Note. Under review; three reviewers recommended oral presentation.*
- **MUSE — multi-slice spatial transcriptomics (co-first author).** Co-first authored a multi-slice joint-analysis method for spatial transcriptomics experiments, complementing single-sample modeling with research on information shared across tissue slices.
- **TargetSage — interpretable biomedical reasoning (co-author).** Responsible for **disease-state verification and multi-target investigation** within an evidence-grounded LLM framework with positive–unlabeled learning. The overall model improved **macro-average Adjusted F1 by 37% relative to the strongest of 10 baselines**, across 15 tasks and 19,032 genes.
- **Perturb-seq and biological foundation-model adaptation.** Used **Perturb-seq data to fine-tune scGPT** in TargetSage and InverseTURB-related research, connecting disease-state context and multi-target reasoning to perturbation-conditioned cellular responses.
- **Spine — context-aware spatial response modeling (first author).** Developed a formulation separating tissue context from perturbation-conditioned response fields. Extended TargetSage reasoning toward spatial perturbation research; manuscript under review.
- **Reproducible evaluation and leakage control.** Constructed in-domain, cross-patient, and cross-platform benchmarks using provenance-aware splits, dataset QC, and training-only molecular reference banks to assess transfer to unseen samples without using held-out molecular profiles.

Technical Skills

Biomedical ML	Computational pathology, H&E/whole-slide imaging, spatial transcriptomics, single-cell genomics, image–omics alignment, cross-sample generalization
Modeling	Multimodal transformers, mixture-of-experts, contrastive learning, retrieval-augmented prediction, scGPT fine-tuning, population/context reliability, LLM reasoning
Engineering	Python, PyTorch, multi-GPU experimentation, large-scale data curation and QC, patient/study/assay-aware evaluation, reproducible benchmarking

Mentoring and Open-Source Leadership

Google Summer of Code (GSoC) **2025 & 2026**
Project Mentor

- Provided **end-to-end mentorship to two students** across the 2025 and 2026 projects, guiding problem formulation, implementation, experiments, and paper development; both projects have code repositories.
- Guided a student to publish **one student-first-author paper at BIBM**.
- Projects: **2026: Omni-ST** — instruction-driven multimodal spatial modeling; **2025: Disentangled Generation and Editing of Pathology Images** — controllable pathology-image generation and editing.

Earlier Research and Industry Experience

Hantao Zhiying *2022 – 2023*
Chief Product Lead. Led AI imaging product definition and delivery across research, engineering, and design.

Multimedia Lab, Tsinghua University *Oct 2021 – Sep 2022*
Researcher. Developed and deployed an audio–visual camera-production system at **China's National Centre for the Performing Arts**.

ByteDance AI Lab *May 2021 – Dec 2021*
AI Product Manager — Full-Time Intern. Incubated **Alchemy** for Fanqie Novel book covers; supported **Chimera** assistant solutions through **XPENG/QCraft** client engagement.

ByteDance — Feishu/Lark *Jun 2020 – Nov 2020*
Data Product — Full-Time Intern. Applied behavior and relevance analysis to enterprise search-product design.

Nolibox / Computational Aesthetics *Mar 2020 – Aug 2020*
Partner and Co-founder. Built automatic logo-generation and font-stroke generation tools for AI-assisted design.

Selected Publications and Manuscripts

Publication links: xenali96.github.io/publications.html.

Published / accepted

1. **Xi Li**, Yaqi Hu, Ziheng Duan, Chaoyang Wang, and Jing Zhang. "STMoE: Multi-Scale Mixture-of-Experts for Single-Cell Gene Expression Prediction from Histology." **CIKM 2026**, accepted. *First author.*

2. **Xi Li**, Yaqi Hu, Ziheng Duan, Diptanshu Sikdar, Yang Liu, and Jing Zhang. "sMMC-22M: A Context-Aware Dataset and Benchmark for Single-Cell Spatial Transcriptomics." **NeurIPS 2026**, accepted poster. *First author.*

3. Ziheng Duan*, **Xi Li***, Zhiqing Xiao, Rex Ying, and Jing Zhang. "MUSE: A Multi-Slice Joint Analysis Method for Spatial Transcriptomics Experiments." **CIKM 2025**. **Co-first author.**

4. Ziheng Duan, Tong Wu, **Xi Li**, Simon Dongbo Sun, Chaoyang Wang, Junhao Liu, Martin Renqiang Min, and Jing Zhang. "TargetSage: Identifying Therapeutic Target Genes with Interpretable and Robust LLM Reasoning." **NeurIPS 2026**, accepted poster.

5. **Xi Li**, Ziheng Duan, Yi Dai, Siwei Xu, and Jing Zhang. "iHAST: Integrating Hybrid Attention for Super-Resolution in Spatial Transcriptomics." **BMVC 2024**. *First author.*

6. Ziheng Duan, **Xi Li**, Zhuoyang Zhang, James Song, and Jing Zhang. "DISCO: A Diffusion Model for Spatial Transcriptomics Data Completion." **ICIP 2025**.

Manuscripts

7. **Xi Li**, Ziheng Duan, Yi Dai, Yang Liu, and Jing Zhang. "Context Is Not a Side Note: A Support-Calibrated Reliability Framework for Biomedical Foundation Models." **KDD 2027**, under review. *First author.*

8. **Xi Li**, Ziheng Duan, Simon Dongbo Sun, Martin Renqiang Min, Yang Liu, and Jing Zhang. "Spine: Disentangling Tissue Context and Perturbation-Conditioned Response Fields in Spatial Genomics." **KDD 2027**, under review. *First author.*

9. Liyun Lin, Shuozhen Bao, **Xi Li**, Isaiah Yim, Dejiang Wang, Gaocai Li, Xiangjun Di, Shujie He, Junyi Xu, Kathryn Miller-Jensen, Jing Zhang, and Rong Fan. "Spatially Indexed Single-Cell Morphology and Transcriptome Profiling by DBiC-seq." **Nature Communications**, under review.

10. "CCR2 Deficiency Attenuates Flow-Induced Carotid Artery Remodeling." **Vascular Pharmacology**, under review. Co-authors include Fu Gao and **Xi Li**; corresponding author: Yang Liu.

11. Fu Gao, Ian Mason, Mingze Dong, Yao Lu, Di Zhang, Xing Lou, **Xi Li**, Irbaz Hameed, Mingyu Yang, Mei Zhong, Markus Krane, Giovanni Ferrari, George Tellides, Yang Liu, Rong Fan, and Arnar Geirsson. "Cross-Species Multi-Omics Identifies a Conserved Fibrogenic Valvular Interstitial Cell Population across Mitral Valve Diseases." **Circulation**, submitted.